

input signal

input_signal — Signal 1

GPS Consistency Score (Location Stability Signal)

Final Formula

$S_{\{i,t\}} = (-)$

Interpretation

This signal measures:

how stable and physically consistent a user’s location history is over time.

It is used in:

$L_i(t)$

Component Breakdown

1. Spatial Variance

$\wedge^2_{\{i,t\}} =$

Variables

- $_i$: set of location samples for user i
- $L_i(t_k)$: location at time t_k
- $\{l\}_i(t)$: weighted mean location

2. Mean Location

$\{l\}_i(t) =$

3. Distance Metric

$| L_i(t_k) - \{l\}_i(t) |$

Meaning

- geographic distance (e.g., haversine)

4. Normalization Parameter

- $_0$: expected spatial deviation scale

Variable Definitions

- $S_{\{i,t\}}$: GPS consistency score
- $\sigma^2_{\{i,t\}}$: location variance
- α : temporal decay
- β : normalization constant

Properties

- bounded in $[0,1]$
- high when movement is realistic
- low when movement is erratic

Behavior Interpretation

Behavior Score stable movement high gradual movement high teleportation low random jumps very low

Why This Signal Matters

Detects:

- GPS spoofing
- bot movement patterns
- unrealistic travel

Usage in System

Feeds into:

$$L_i(t) = w_1 S_{\{i,t\}} +$$

Pseudo Insight

Real humans move smoothly, attackers jump

input_signal — Signal 2

Speed Plausibility Score (Physical Movement Constraint)

Final Formula

$$S_{\{i,t\}}(i,t) = (-)$$

Interpretation

This signal measures:

whether the user's movement speed is physically plausible over time.

It directly detects:

- teleportation
- GPS spoofing
- impossible travel speeds

Component Breakdown

1. Instantaneous Speed Between Consecutive Points

$v_k =$

Variables

- $l_i(t_k)$: location at time t_k
- $d(,)$: geographic distance (e.g., haversine)
- $t_k - t_{k-1}$: time difference

2. Maximum Observed Speed

$v_{\{i,t\}} = \max_k v_k$

Meaning

- worst-case movement speed
- captures extreme anomalies

3. Normalization Parameter

- v_0 : maximum plausible speed

Typical Values

Walking $\rightarrow \sim 1.5$ m/s Vehicle $\rightarrow \sim 30$ m/s Airplane $\rightarrow \sim 250$ m/s

Choose v_0 based on your system assumption.

4. Exponential Decay Mapping

(-)

Meaning

- small speeds $\rightarrow$ near 1
- large speeds $\rightarrow$ rapidly decreases

Variable Definitions

- $S_{\{i,t\}}$: speed plausibility score
- v_k : instantaneous speed
- $v_{\{i,t\}}$: maximum observed speed
- v_0 : speed threshold

Properties

- bounded in $[0,1]$
- sensitive to extreme values
- robust to small noise

Behavior Interpretation

Behavior Score normal movement high fast but realistic moderate teleportation near 0 extreme jumps 0

Why Max Speed Is Used

Using:

$$v_{\{}}$$

instead of average ensures:

even a single impossible jump is detected

Usage in System

Feeds into:

$$L_i(t) = w_1 S_{\{}} + w_2 S_{\{}} +$$

Pseudo Insight

One impossible jump is enough to lose trust

input_signal — Signal 3

Location Source Reliability Score (Sensor Trustworthiness)

Final Formula

$$S_{\{}}(i,t) = \sum_s q_s(i,t)$$

Constraint

$$\sum_s q_s = 1$$

Interpretation

This signal measures:

how trustworthy the source of location data is for user i at time t .

It captures:

- sensor type reliability
- signal integrity
- spoofing likelihood

Component Breakdown

1. Set of Location Sources

$$= \{i_1, i_2, i_3\}$$

2. Source Quality Score

$q_s(i,t)$

Meaning

Represents reliability of source s for user i

Typical Values

Source q_s GPS (high accuracy) high WiFi triangulation medium Cellular lower IP-based lowest

3. Source Weight

w_s

Meaning

- importance assigned to each source
- system-level configuration

Example

GPS $\rightarrow$ 0.5 WiFi $\rightarrow$ 0.3 Cellular $\rightarrow$ 0.15 IP $\rightarrow$ 0.05

4. Combined Score

$S_{\{i,t\}}$

Meaning

- weighted average of all active sources

Enhancement: Consistency Across Sources

Optional refinement:

$q_s(i,t) = (-)$

Meaning

- penalizes disagreement between sources

Variable Definitions

- $S_{\{i,t\}}$: source reliability
- $q_s(i,t)$: quality of source s
- w_s : weight of source
- l_s : location from source s

Properties

- bounded in $[0,1]$
- increases with high-quality sensors

- decreases with inconsistent signals

Behavior Interpretation

Scenario Score GPS + WiFi consistent high only IP low conflicting sources reduced spoofed GPS reduced

Why This Signal Matters

Detects:

- fake GPS apps
- IP masking
- inconsistent sensor data

Usage in System

Feeds into:

$$L_i(t) = w_1 S_{\{i\}} + w_2 S_{\{i\}} + w_3 S_{\{i\}} +$$

Pseudo Insight

Not all location signals are equal

input_signal — Signal 4

Movement Continuity Score (Trajectory Smoothness)

Final Formula

$$S_{\{i\}}(i,t) = (-)$$

Interpretation

This signal measures:

how smooth and physically continuous a user's movement trajectory is over time.

It captures:

- abrupt direction changes
- zig-zag or random movement
- non-human trajectory patterns

Component Breakdown

1. Direction Vectors

$$k = L_i(t_k) - L_i(t_{k-1})$$

Meaning

- displacement vector between consecutive positions

2. Angle Between Consecutive Movements

$$_k = ()$$

Meaning

- change in direction between steps

3. Trajectory Angular Variance

$$A_i(t) =$$

Meaning

- weighted measure of direction instability

4. Normalization Parameter

- A_0 : expected angular variation for normal movement

5. Final Score

$$S_{\{ \}}(i,t)$$

Meaning

- smooth trajectory $\rightarrow$ high score
- erratic movement $\rightarrow$ low score

Variable Definitions

- $S_{\{ \}}(i,t)$: continuity score
- $_k$: angle between movements
- $A_i(t)$: angular variance
- $_L$: time decay
- A_0 : normalization constant

Properties

- bounded in $[0,1]$
- sensitive to rapid direction changes
- robust to small noise

Behavior Interpretation

Movement Type Score straight path high gradual turns high zig-zag low random jumps very low

Why This Signal Matters

Detects:

- scripted/bot movement
- GPS spoofing patterns

- non-physical trajectories

Usage in System

Feeds into:

$$L_i(t) = w_1 S_{\{i\}} + w_2 S_{\{i\}} + w_3 S_{\{i\}} + w_4 S_{\{i\}} +$$

Pseudo Insight

Humans move smoothly, bots move erratically

input_signal — Signal 5

Interaction Rate Deviation (Burst Detection Signal)

Final Formula

$$S_{\{i\}}(i,t) =$$

Optional Bounded Form (Recommended for Stability)

$$_{\{i\}}(i,t) = 1 - (-)$$

Interpretation

This signal measures:

how much a user's current activity deviates from their normal behavior.

It detects:

- sudden bursts
- spam activity
- coordinated attack behavior

Component Breakdown

1. Short-Term Activity

$$V_i^{\{window\}}(t) = \sum_{k: t_k} 1$$

Meaning

- number of actions in recent time window

2. Baseline Activity Rate

$$_i^{\{baseline\}}(t) =$$

Simplified Practical Form

baseline = average interactions per unit time (historical)

3. Ratio

Meaning

- compares current activity vs normal behavior

4. Exponential Bounding

1 - (-x)

Purpose

- prevents unbounded growth
- stabilizes anomaly contribution

Variable Definitions

- $S_{\{i,t\}}$: burst signal
- $V_{i^{\{window\}}(t)}$: recent activity count
- $_i^{\{baseline\}}(t)$: long-term average rate
- t : time window
- $_r$: decay rate
- ϵ : small constant

Properties

- $S_{\{i,t\}}$ (unbounded raw form)
- bounded version $_i^{\{baseline\}}(t)$
- sensitive to sudden spikes

Behavior Interpretation

Behavior Signal normal activity ≈ 1 mild increase > 1 strong burst $\gg 1$ no activity ≈ 0

Why This Signal Matters

Detects:

- bot spamming
- coordinated voting bursts
- artificial amplification

Usage in System

Feeds into anomaly component:

$$D_1(i,t) = _i^{\{baseline\}}(t)$$

Pseudo Insight

Humans are consistent, attackers spike

input_signal — Signal 6

Action Entropy Score (Behavioral Diversity Signal)

Final Formula

$$S_{\{i\}}(t) =$$

Where

$$H_i(t) = - \sum_{a \in A} p_i(a,t) \log_2 p_i(a,t)$$

Interpretation

This signal measures:

how diverse and natural a user's actions are over time.

It captures:

- repetitive behavior
- scripted actions
- lack of variability

Component Breakdown

1. Action Set

Meaning

Set of possible user actions, for example:

{vote_up, vote_down, post, scroll, view, share}

2. Action Probability Distribution

$$p_i(a,t) =$$

Meaning

- probability that user performs action a
- time-decayed frequency

3. Entropy Calculation

$$H_i(t)$$

Meaning

- measures randomness/diversity
- higher $\rightarrow$ more varied behavior

4. Normalization

||

Meaning

- maximum possible entropy
- ensures output is bounded

5. Final Score

$S_{\{i,t\}}$

Variable Definitions

- $S_{\{i,t\}}$: entropy score
- $H_i(t)$: action entropy
- $p_i(a,t)$: probability of action
- $_r$: decay parameter
- $||$: number of actions

Properties

- bounded in $[0,1]$
- high for diverse behavior
- low for repetitive behavior

Behavior Interpretation

Behavior Score varied actions high moderate diversity medium repetitive voting only low identical actions near 0

Why This Signal Matters

Detects:

- bots repeating same action
- coordinated scripts
- non-human interaction patterns

Usage in System

Feeds into anomaly:

$$D_2(i,t) = 1 - S_{\{i,t\}}$$

Key Insight

Pseudo Insight

Humans behave variably, bots repeat

input_signal — Signal 7

Consensus Alignment Score (Correctness Consistency Signal)

Final Formula

$$S_{\{i\}}(t) =$$

Interpretation

This signal measures:

how consistently user i 's actions align with the final verified outcomes over time.

It captures:

- correctness of behavior
- long-term alignment with truth
- deviation from consensus

Component Breakdown

1. Interaction Set

I_i

Meaning

- all interactions performed by user i

2. Indicator Function

$$(s_k = y_{\{j\}_k})$$

Meaning

- equals 1 if:
 - user vote s_k matches final truth $y_{\{j\}_k}$
- equals 0 otherwise

3. Time Decay

$$e^{-r(t - t_k)}$$

Meaning

- recent correctness is more important
- older actions fade

4. Normalization

$$\sum_k e^{-r(t - t_k)}$$

Meaning

- total weighted interactions

5. Final Score

$S_{\{i,t\}}$

Variable Definitions

- $S_{\{i,t\}}$: consensus alignment score
- s_k : user vote
- $y_{\{j_k\}}$: ground truth label of post
- $_r$: decay parameter
- t_k : interaction timestamp
- ϵ : numerical stability constant

Properties

- bounded in $[0,1]$
- increases with correct behavior
- decreases with incorrect behavior

Behavior Interpretation

Behavior Score mostly correct high mixed accuracy medium mostly incorrect low adversarial near 0

Why This Signal Matters

Captures:

- long-term correctness
- deviation from truth
- reliability consistency

Usage in System

Feeds into:

Reliability (implicitly)

$_i,; _i$

Anomaly

$$D_3(i,t) = 1 - S_{\{i,t\}}$$

Important Note

This signal requires:

ground truth y_j available (delayed)

So it is:

- retrospective

- not available immediately

Pseudo Insight

Consistently wrong behavior is a strong anomaly signal

input_signal — Signal 8

Coordination Similarity Score (User–User Behavioral Correlation)

Final Formula

$$S_{\{i,t\}} = \sum_j (i) \text{ Sim}(i,j,t)$$

Where

$$\text{Sim}(i,j,t) =$$

Interpretation

This signal measures:

how strongly user i behaves in coordination with other users.

It detects:

- synchronized voting
- coordinated attacks
- bot clusters

Component Breakdown

1. Neighbor Set

(i)

Meaning

- users who have interacted with the same posts as user i

2. Shared Interaction Set

$\{i,j\}$

Meaning

- set of posts where both users i and j interacted

3. Vote Agreement Indicator

$$(s_{\{i,k\}} = s_{\{j,k\}})$$

Meaning

- 1 if both users gave same vote
- 0 otherwise

4. Temporal Alignment Factor

(-)

Meaning

- high when actions occur close in time
- low when actions are far apart

5. Averaging

Meaning

- normalizes similarity across shared interactions

6. Maximum Selection

$\max_j (i)$

Meaning

- focuses on strongest coordination partner
- captures worst-case coordination

Variable Definitions

- $S_{ij}(t)$: coordination score
- $\text{Sim}(i,j,t)$: similarity between users
- $s_{i,k}, s_{j,k}$: votes
- $t_{i,k}, t_{j,k}$: timestamps
- α : time sensitivity parameter

Properties

- bounded in $[0,1]$
- increases with:
 - agreement
 - synchronization
- robust to random overlap

Behavior Interpretation

Behavior Score independent users low occasional agreement moderate synchronized voting high bot cluster near 1

Why Max Operator Is Used

Using:

$_j \text{ Sim}(i,j)$

ensures:

even a single strong coordination link is detected

Why This Signal Matters

Detects:

- coordinated misinformation campaigns
- bot networks
- group manipulation

Usage in System

Feeds into anomaly:

$D_4(i,t) = S_{\{i,t\}}$

Pseudo Insight

Acting together too perfectly is suspicious

input_signal — Signal 9

Location Inconsistency Score (Teleportation / Plausibility Violation)

Final Formula

$S_{\{i,t\}} =$

Interpretation

This signal measures:

how frequently a user's location transitions violate physical plausibility.

It captures:

- teleportation behavior
- long-distance jumps in short time
- spoofed location traces

Component Breakdown

1. Movement Set

$i = \{t_{k-1}, t_k\}$

Meaning

- consecutive location transitions for user i

2. Instantaneous Speed

$v_k =$

Meaning

- speed required to move between two points

3. Plausibility Indicator

$$I_k = (v_k > v_{\max})$$

Meaning

- 1 if speed exceeds maximum plausible threshold
- 0 otherwise

4. Count of Implausible Movements

$$N_{\text{impl}}(i, t) = \sum_k I_k e^{-\lambda (t - t_k)}$$

5. Total Movement Count

$$N_{\text{total}}(i, t) = \sum_k e^{-\lambda (t - t_k)}$$

6. Final Ratio

$$S_{\text{impl}}(i, t)$$

Meaning

- fraction of implausible movements

Variable Definitions

- $S_{\text{impl}}(i, t)$: inconsistency score
- v_k : movement speed
- $v_{\max}$: maximum plausible speed
- λ : time decay
- $d(i, j)$: geographic distance
- ϵ : small constant

Properties

- bounded in $[0, 1]$
- increases with abnormal movement frequency
- robust to occasional noise

Behavior Interpretation

Behavior Score consistent movement near 0 occasional anomaly low frequent jumps high teleportation pattern near 1

Important Relation

This signal complements:

- Speed Plausibility (Signal 2) → detects extreme single event
- Location Inconsistency (this) → detects repeated violations

Why This Signal Matters

Detects:

- repeated GPS spoofing
- bot-generated location traces
- coordinated location fraud

Usage in System

Feeds into anomaly:

$$D_5(i,t) = S_{\{i\}}(i,t)$$

Pseudo Insight

Occasional anomaly is noise, repeated anomaly is intent

input_signal — Signal 10

Human Navigation Deviation Score (Behavioral Mobility Model)

Final Formula

$$S_{\{i\}}(i,t) = 1 - (-)$$

Where

$$D_{\{i\}}(i,t) = \{i\} d\{P_i(t), P\{i\}\}$$

Interpretation

This signal measures:

how much a user's movement behavior deviates from typical human navigation patterns.

It captures:

- unrealistic travel sequences
- abnormal mobility patterns
- non-human trajectory behavior

Component Breakdown

1. Trajectory Segments

P_i

Meaning

- set of trajectory segments (e.g., sequences of locations over time)

2. User Mobility Distribution

$P_i()$

Meaning

- probability distribution over trajectory features for user i

Features may include:

step length turn angles pause duration movement frequency

3. Human Reference Distribution

$P_{\{ \}}()$

Meaning

- learned distribution from real human mobility data

4. Distribution Distance

$d_{\{ \}}(P_i, P_{\{ \}})$

Options

KL divergence Wasserstein distance Jensen-Shannon divergence

Recommended

$d_{\{ \}} = D_{\{JS\}}(P_i, P_{\{ \}})$

5. Averaging Across Segments

Meaning

- average deviation across movement patterns

6. Exponential Mapping

$1 - (-x/)$

Meaning

- smooth mapping to $[0,1]$
- controlled sensitivity

Variable Definitions

- $S_{\{ \}}(i,t)$: navigation deviation score

- $D_{\{i,t\}}$: deviation magnitude
- α : scaling parameter
- P_i : user mobility distribution
- $P_{\{i\}}$: reference distribution

Properties

- bounded in $[0,1]$
- increases with behavioral abnormality
- robust to noise

Behavior Interpretation

Behavior Score normal human mobility low slightly unusual moderate highly irregular high synthetic movement near 1

Why This Signal Matters

Detects:

- synthetic trajectories
- scripted location behavior
- advanced spoofing (beyond simple jumps)

Difference from Other Signals

Signal Detects GPS consistency variance speed extreme jumps continuity direction changes navigation (this) distribution-level behavior

Usage in System

Feeds into anomaly ML features:

$$r_{\{i,t\}} = S_{\{i,t\}}$$

Pseudo Insight

Not just where you go, but how you move matters

Status So Far

You now have 10 core signals, covering:

Location Trust Signals

1. GPS consistency
2. Speed plausibility
3. Source reliability
4. Movement continuity
5. Location inconsistency
6. Navigation deviation

Behavioral Signals

7. Interaction rate deviation
8. Action entropy
9. Consensus alignment
10. Coordination similarity

What Remains (Final Signals)

To complete system:

- device fingerprint reliability
- IP consistency score
- session continuity
- vote timing irregularity

input_signal — Signal 11

Device Fingerprint Consistency Score (Identity Stability Signal)

Final Formula

$$S_{\{i,t\}} = (-)$$

Where

$$H_{\{i,t\}} = - \sum_{d \in \mathcal{D}_i} p_i(d,t) \log p_i(d,t)$$

Interpretation

This signal measures:

how consistent a user's device usage is over time.

It captures:

- multiple device switching
- bot/device rotation
- identity instability

Component Breakdown

1. Device Set

$\mathcal{D}_i$

Meaning

- set of devices used by user i

Device fingerprint may include:

browser + OS hardware signature screen resolution user-agent hash

2. Device Probability Distribution

$$p_i(d,t) =$$

Meaning

- probability of user using device d
- time-decayed frequency

3. Device Entropy

$$H_{\{i\}}(i,t)$$

Meaning

- measures diversity of device usage
- higher entropy $\rightarrow$ more switching

4. Normalization

$$|_{\{i\}}$$

Meaning

- maximum entropy possible

5. Exponential Mapping

$$(-x)$$

Meaning

- converts entropy to consistency
- high entropy $\rightarrow$ low score

Variable Definitions

- $S_{\{i\}}(i,t)$: device consistency score
- $H_{\{i\}}$: device entropy
- $p_i(d,t)$: device usage probability
- $_d$: decay parameter
- $_i$: device set

Properties

- bounded in $[0,1]$
- high for consistent device usage
- low for frequent switching

Behavior Interpretation

Behavior Score single device high few devices moderate frequent switching low rotating devices (botnet) near 0

Why This Signal Matters

Detects:

- account sharing
- bot farms
- identity spoofing

Usage in System

Feeds into anomaly ML features:

$r_{\{i,t\}} = S_{\{i,t\}}$

Pseudo Insight

Real users are stable, bots rotate identities

input_signal — Signal 12

IP Consistency Score (Network-Level Stability Signal)

Final Formula

$S_{\{i,t\}} = (-) (-)$

Interpretation

This signal measures:

how stable and geographically consistent a user’s network identity (IP) is over time.

It captures:

- IP hopping
- VPN/proxy usage
- mismatch between IP location and GPS

Component Breakdown

1. IP Set

$_i$

Meaning

- set of distinct IP addresses used by user i

2. IP Usage Distribution

$p_i(ip,t) =$

Meaning

- probability of using a specific IP
- time-decayed

3. IP Entropy

$$H_{\{i\}}(i,t) = - \sum_{ip} p_i(ip,t) \log p_i(ip,t)$$

Meaning

- measures IP switching behavior
- higher entropy $\rightarrow$ more switching

4. IP-Based Location Variance

$$\sigma^2_{\{i\}}(i,t) =$$

Variables

- $l_{\{ip\}}(t_k)$: location derived from IP
- $\bar{l}_{\{ip\}}(t)$: mean IP location

5. Mean IP Location

$$\bar{l}_{\{ip\}}(t) =$$

6. Combined Score

Two factors:

1. IP switching (entropy)
2. IP geographic consistency (variance)

Variable Definitions

- $S_{\{i\}}(i,t)$: IP consistency score
- $H_{\{i\}}$: entropy of IP usage
- $\sigma^2_{\{i\}}$: geographic variance
- $\alpha_{\{ip\}}$: decay parameter
- $\beta_{\{ip\}}$: normalization constant

Properties

- bounded in $[0,1]$
- decreases with:
 - frequent IP switching
 - inconsistent geographic origin

Behavior Interpretation

Behavior Score stable IP high few IPs (normal mobility) moderate VPN switching low global IP hopping near 0

Why This Signal Matters

Detects:

- VPN/proxy abuse
- bot networks using rotating IPs
- mismatch between physical and network location

Relation to Other Signals

Signal Layer GPS consistency device/location IP consistency network layer device fingerprint identity layer

Usage in System

Feeds into anomaly ML features:

$$r_{\{i,t\}} = S_{\{i,t\}}$$

Pseudo Insight

Real users stay locally consistent, attackers jump networks

input_signal — Signal 13

Session Continuity Score (Human Activity Pattern Signal)

Final Formula

$$S_{\{i,t\}} = (-)$$

Where

$$D_{\{i,t\}} = +$$

Interpretation

This signal measures:

how closely a user's session behavior matches typical human activity patterns.

It captures:

- unrealistically long sessions (bots running continuously)
- extremely short fragmented sessions
- abnormal session timing

Component Breakdown

1. Session Definition

A session is defined as:

continuous activity with $\text{gap} \leq T_{\text{gap}}$

Meaning

- if time between actions exceeds T_{gap} , a new session starts

2. Session Duration Set

$$_i = \{d_1, d_2, \dots, d_n\}$$

Meaning

- durations of all sessions for user i

3. Mean Session Duration

$$\bar{d}_i =$$

4. Session Duration Variance

$$\sigma_i^2 =$$

5. Human Reference Values

- $\bar{d}_{\text{human}}$: average human session duration
- σ_{human} : expected variation

Example

$$\mu_{\text{human}} \approx 5\text{--}20 \text{ minutes } \sigma_{\text{human}} \approx \text{moderate variability}$$

6. Deviation Measure

$$D_i(t)$$

Meaning

- normalized difference from human baseline

7. Final Score

$$S_i(t)$$

Meaning

- human-like sessions $\rightarrow$ high score
- abnormal sessions $\rightarrow$ low score

Variable Definitions

- $S_i(t)$: session continuity score
- d_k : session duration
- α : decay parameter
- β : sensitivity constant

Properties

- bounded in [0,1]
- penalizes extreme behavior
- robust to normal variation

Behavior Interpretation

Behavior Score normal session patterns high slightly irregular medium continuous 24/7 activity low extremely fragmented low

Why This Signal Matters

Detects:

- bots running continuously
- scripted periodic behavior
- unnatural usage patterns

Usage in System

Feeds into anomaly ML features:

$$r_{\{i,t\}} = S_{\{i,t\}}$$

Pseudo Insight

Humans take breaks, bots don't

input_signal — Signal 14

Vote Timing Irregularity Score (Temporal Behavior Signal)

Final Formula

$$S_{\{i,t\}} = (-) (-)$$

Interpretation

This signal measures:

how natural the timing of a user's actions is.

It captures:

- perfectly regular (bot-like) timing
- synchronized bursts
- unnatural reaction intervals

Component Breakdown

1. Inter-Event Time Differences

$$t_k = t_k - t_{\{k-1\}}$$

Meaning

- time gap between consecutive actions

2. Variance of Time Gaps

$$\hat{2}_{\{t\}}(i,t) =$$

3. Mean Time Gap

$$\{t\} =$$

Meaning

- average delay between user actions

4. Burstiness Measure

$$B_i(t) =$$

Where

- $_t$: small time bins (e.g., 1 minute windows)
- n_k : number of actions in bin k
- $\{n\}_i$: average actions per bin

Meaning

- detects sudden spikes in activity

5. Final Combination

Two penalties:

1. low variance $\rightarrow$ too regular (bot-like)
2. high burstiness $\rightarrow$ coordinated/spam behavior

Variable Definitions

- $S_{\{t\}}(i,t)$: timing score
- t_k : time gaps
- $\hat{2}_{\{t\}}$: variance of gaps
- $B_i(t)$: burstiness
- $_t$: decay parameter
- $_t, B_0$: normalization constants

Properties

- bounded in $[0,1]$
- penalizes:
 - overly regular timing
 - extreme bursts

Behavior Interpretation

Behavior Score natural irregular timing high slightly structured medium perfectly periodic low synchronized bursts low

Why This Is Fully Defined (No Abstraction)

Everything is derived from:

timestamps only

No need for:

- external datasets
- learned distributions
- hidden models

Usage in System

Feeds into anomaly ML:

$$r_{\{i,t\}} = S_{\{i,t\}}$$

Now Addressing Your Concern Directly

You were correct—some earlier signals can be abstract if implemented loosely. Here is the correction:

Signals That MUST Be Implemented Carefully

1. Navigation Deviation (Signal 10)

Originally abstract → now should be implemented using:

histograms of: step length angle pause time compare via Jensen-Shannon divergence

No black-box distribution.

2. Category / Keyword Scores

Must be defined as:

keyword → dictionary or TF-IDF weights category → classifier with explicit output $\in [0,1]$

3. Source Reliability

Must not be static—use:

agreement between sources (GPS vs IP)

Final Status

You now have 14 fully grounded signals, each:

- computable from raw data

- bounded or normalized
- tied directly to formulas

Critical Insight

Your system is now:

signal-complete + mathematically closed

No variable depends on undefined inputs anymore.

Short answer: you are ~95% complete, but not fully closed yet. There are still a few variables that are either:

- implicitly assumed
- environment-dependent
- or not explicitly mapped to raw data collection

I'll go through this surgically, not vaguely.

1. What is Fully Defined (No Gaps)

These are safe and computable from raw logs:

✓ User interaction signals

- timestamps t_k
- votes s_k
- post IDs
- interaction counts

✓ Behavioral signals

- $S_{\{}}$
- $S_{\{}}$
- $S_{\{}}$
- $S_{\{}}$

All depend only on:

user_id, action_type, timestamp, vote

No ambiguity.

✓ Graph + trust

- adjacency $A_{\{ij\}}$
- trust propagation

Depends only on:

co-interactions + timestamps + votes

Fully computable.

✓ Post-level variables

- $N_j(t)$, $S_j(t)$, C_{Bayes} , Var_j

All derived from:

user weights + interactions

No missing input.

2. Variables That Are Defined BUT Require Explicit Data Collection

These are not undefined—but you must explicitly collect them.

A. Location Data

Variables:

- $L_i(t)$ (user location)
- L_j (post location)

Required Inputs

GPS coordinates (lat, lon, timestamp) OR IP-based geo fallback

Risk

If you don't collect location:

entire spatial system (Prox, L_i , alerts) breaks

B. Device Fingerprint

Variables:

- $_i$, device IDs

Required Inputs

user-agent device_id (hashed) OS + browser

Risk

Without it:

Signal 11 becomes unusable

C. IP Data

Variables:

- ip_k , L_{ip}

Required Inputs

IP address per request Geo-IP mapping

Risk

Without it:

network-level anomaly detection is gone

D. Content Features

Variables:

- p_j (text)
- keywords (w)
- category Cat_j

Required Inputs

raw post text

Additional Requirement

You must define:

keyword scoring method category classifier

3. Variables That Still Need Explicit Implementation Decisions

These are not undefined, but not concretely specified yet.

A. Ground Truth y_j

Used in:

$S_{\{j\}}; _i; _i$

Problem

You never defined:

where truth comes from

Options

manual moderation fact-check APIs delayed consensus (majority after stabilization)

Without this:

- reliability becomes weak
- anomaly loses strength

B. Human Reference Constants

Used in:

- navigation deviation
- session continuity

Variables:

- $_{}^{\{}}$
- $_{}^{\{}}$
- $P_{}^{\{}}$

Issue

These are:

not raw inputs → must be estimated

Fix

Define:

compute from platform-wide statistics

C. Hyperparameters

You defined them, but they are not “collected”:

lambda, sigma, tau, alpha, beta, etc.

These require:

- tuning
- validation
- simulation

D. Similarity Function $\text{Sim}(j,k)$ (Memory)

You wrote:

$\text{Sim}(j,k)$

But not concretely:

- cosine over embeddings?
- TF-IDF?

Fix

Define explicitly:

$\text{Sim} = \text{cosine}(\text{embedding_j}, \text{embedding_k})$

4. Subtle Missing Link (Important)

User Identity Mapping

You assume:

i =

But in reality:

You must define:

user_id ↔ device ↔ session ↔ IP

Otherwise:

- signals conflict
- spoofing becomes trivial

5. Final Verdict

Do you have all inputs?

✓ YES (core system)

You can compute:

- credibility
- anomaly
- propagation
- alerts

⚠ PARTIAL (requires infrastructure)

You still need to collect explicitly:

location (GPS/IP) device fingerprint IP logs post text timestamps (high precision)

⚠ NOT DEFINED (external/system-level)

You must still decide:

ground truth source human baseline estimation embedding model for similarity

6. Clean Classification

Fully Ready

- mathematical formulas
- algorithms
- signal definitions

Needs Data Engineering

- location

- device
- IP

Needs System Policy

- truth labeling
- moderation strategy

7. Final Insight (Important)

You are not blocked by math anymore.

You are now at:

data collection + system design phase

8. Direct Answer

Do we have any superficial variable left?

Yes, but only 3 categories:

1. Ground truth y_j (not defined source)
2. Human baseline stats (must be learned)
3. Content similarity definition (must be fixed)

Everything else is:

fully computable from user data